SALIFORT MOTORS

Employee retention | Executive summary

Sebastián Madrigal López · Google Advanced Data Analytics capstone project

0.968	90.5%	88.0%	2,998
ROC AUC	PRECISION	RECALL	TEST ROWS

Reduced-input random forest · Positive class: left · Default threshold: 0.5

ISSUE / PROBLEM

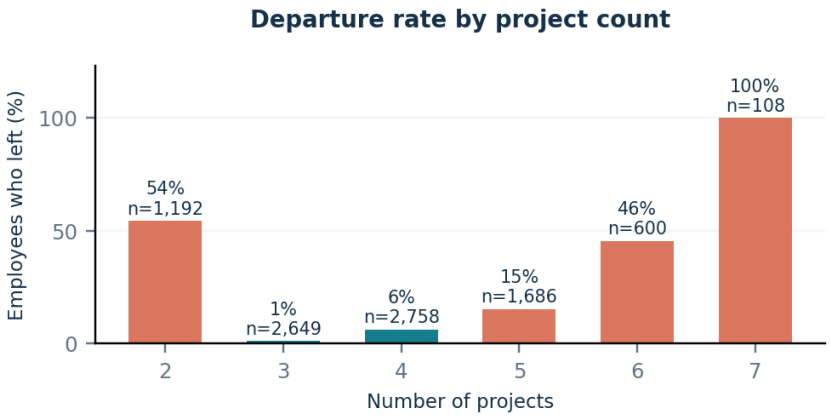
HR seeks to understand employee departures and identify priorities for retention research. The analysis examines workload, tenure, satisfaction and evaluations in the Salifort Motors case study.

RESPONSE

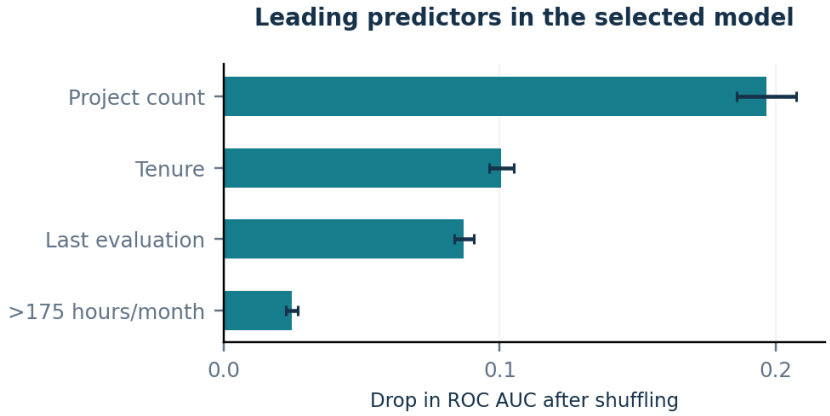
Analyzed 11,991 records after removing 3,008 exact repeats. Compared decision trees and random forests with full and reduced inputs. Selected the reduced-input forest by four-fold CV ROC AUC and evaluated it on 2,998 test rows.

IMPACT

The selected model detected 438 of 498 departures, with 46 false positives. Project count, tenure and evaluation scores provide priorities for HR follow-up. Retention effects require prospective evaluation.



Training data. Labels show departure rates and group sizes.



Permutation importance on test rows. Error bars: standard deviation across 5 shuffles.

INSIGHTS / NEXT STEPS

Review project allocation at both workload extremes, career progression and evaluation practices. Confirm predictor timing and exit type, validate on a later cohort, and pilot a support intervention with measurable employee outcomes.

Limits: Internal evaluation on an educational dataset; undocumented dates and predictor timing; repeated rows treated as redundant. Generalization requires validation on a later cohort.

Source: supplied HR capstone data. Methods and complete results accompany this summary.